Anytime-Valid Conjoint Analysis: A European Asylum-Seeker Case Study

Supplementary case study for *Anytime-Valid Inference in Conjoint Experiments* (Molitor, Gosciak, and Lindon, 2026)

July 14, 2026

1 Introduction

The parent paper establishes anytime-valid e-values and confidence sequences (CSs) for the linear-probability ("AMCE") model of conjoint analysis, and demonstrates their properties through simulation. This note is a second real-data case study, complementing the digital-camera case study with a large-scale political-science conjoint experiment: a forced-choice survey of European attitudes toward asylum seekers. Relative to the camera study (332 respondents, 10 AMCEs from mostly binary attributes), this dataset is a stress test of scale and complexity – 18,021 respondents and nine multi-level categorical attributes, producing 27 AMCEs – and it surfaces real features that the smaller camera dataset did not: some coefficients' e-values grow large enough to overflow floating point, not every AMCE ends up distinguishable from zero even at the full sample, and the conventional and anytime-valid conclusions concretely disagree for a handful of attribute levels. All of these are reported below exactly as found, verified directly against the generated data rather than assumed.

All code, data, and figures for this case study live in `asylum_case_study/` and can be regenerated with `run_all.sh`; see the accompanying `README.md` for exact reproduction steps.

2 Data

We use the 2016 wave of the conjoint experiment from Bansak, Hainmueller, and Hangartner (2016), fielded across 15 European countries. Respondents were shown pairs of hypothetical asylum-seeker profiles and, in each of 5 tasks, forced to choose which of the two profiles they would prefer to see admitted. Each profile was described by nine randomized attributes. The specific CSV analyzed here was released as part of the replication archive for a related follow-up study, Bansak, Hainmueller, and Hangartner (2023), via Harvard Dataverse (see References); that archive is the provenance of this file, while the study design and this wave of data originate in the 2016 paper.

The raw file contains 180,300 rows (18,030 respondents $\times$ 5 tasks $\times$ 2 profiles). Nine respondents (90 rows) have every one of the nine attribute columns missing – corrupted, incomplete records – and are dropped; no other rows have any missing attribute. After cleaning, 18,021 respondents remain, contributing 180,210 rows in total (10 rows per respondent: 5 tasks $\times$ 2 profiles), with every respondent's rows verified to be complete and every task a genuine forced choice (exactly one of the two profiles preferred). We treat each respondent as a cluster, exactly as in the simulations in the main paper and in the camera case study.

The nine attributes, converted to factors before fitting, are:

- **Testimony consistency** (`cconsist`): Major inconsistencies (baseline), Minor inconsistencies, No inconsistencies
- **Gender** (`cgender`): Female (baseline), Male
- **Country of origin** (`corigin`): Afghanistan (baseline), Eritrea, Iraq, Kosovo, Pakistan, Syria, Ukraine
- **Age** (`cage`): 21 Years (baseline), 38 Years, 62 Years
- **Previous occupation** (`cjob`): Accountant (baseline), Cleaner, Doctor, Farmer, Teacher, Unemployed
- **Vulnerability** (`cvulner`): Handicapped (baseline), No surviving family, None, PTSD, Victim of torture
- **Reason for migrating** (`creason`): Economic opportunities (baseline), Ethnic persecution, Political persecution, Religious persecution
- **Religion** (`creligion`): Agnostic (baseline), Christian, Muslim
- **Language skills** (`clang`): Broken (baseline), Fluent, None

The first-listed (alphabetically first) level of each attribute is the reference level against which its AMCEs are measured, R's default `factor()` convention. We fit

$$\texttt{pref} \sim \texttt{cconsist} + \texttt{cgender} + \texttt{corigin} + \texttt{cage} + \texttt{cjob} + \texttt{cvulner} + \texttt{creason} + \texttt{creligion} + \texttt{clang}$$

as a linear probability model via `feols()`, clustering standard errors by respondent (`ResponseId`). Each of the 27 non-intercept coefficients is an AMCE: the average marginal effect of that attribute level, relative to its attribute's baseline, on the probability a profile is preferred.

3 Method

Point estimates and cluster-robust ("CRV1") standard errors come directly from `feols()`. Anytime-valid p-values and confidence-sequence radii are then computed from those same estimates using the e-value construction of the parent paper, with the reference degrees of freedom and the process index both set to $G - 1$ and G respectively, where G is the number of clusters (respondents) observed so far – not the row count – since clusters, not rows, are the unit that accumulates sequentially here.

The tuning parameter g, which controls the shape of the confidence sequence, is fixed once in advance at the value that minimizes the CS radius at the dataset's actual final size, $G = 18,021$. This gives $g^{\star} = 2193.92$. Critically, $g^{\star}$ is computed once from the final sample size and held fixed throughout sequential monitoring below; re-optimizing g as data accumulate would invalidate the anytime-valid guarantee, since g would then depend on how the monitoring unfolds. All significance levels below use $\alpha = 0.05$, so the e-value rejection threshold is $1/\alpha = 20$.

Refitting at every single respondent, as the smaller camera case study does, is unnecessary here and costly at this scale: with 18,021 respondents that would mean roughly that many `feols()` calls plus as many `dplyr::filter()` passes over a 180,000-row frame. A short timing test (fits spread across the full range of G, including the filtering step) measured roughly 0.02-0.13 seconds per fit, growing mildly with G. We therefore step the sequential refit by 100 respondents, giving 182 total fits from $G = 5$ up to $G = 18,021$ (the final value is always included, so the last sequential fit exactly reproduces the final fit below) – comfortably within the targeted 150-400 refits, and measured at 12 seconds of actual runtime, well inside a 3-minute budget.

With 27 non-baseline coefficients across nine attributes, faceting every coefficient in the two over-time figures below (confidence sequences and e-values, Figures 1 and 2) would be unreadable. For those two figures only, we curate exactly one representative coefficient per attribute, chosen by an objective, pre-specifiable rule: **the non-baseline level with the largest $|t|$ statistic in the final full-sample fit, within each attribute**. All 27 non-intercept coefficients are retained in the underlying data (`final_comparison.csv`, `sequential_av_estimates.csv`) and in Figure 3; only Figures 1 and 2 are curated for readability. The nine coefficients this rule selects are:

Attribute	Level	Estimate	t statistic
Testimony consistency	No inconsistencies	0.112	38.1
Gender	Male	-0.059	-25.6
Country of origin	Syria	0.019	4.6
Age	62 Years	-0.063	-21.4
Previous occupation	Unemployed	-0.086	-21.5
Vulnerability	Victim of torture	0.085	23.0
Reason for migrating	Ethnic persecution	0.154	45.6
Religion	Muslim	-0.068	-23.7
Language skills	None	-0.060	-21.2

4 Results

4.1 Confidence sequences over time

Figure 1 shows each curated attribute's confidence sequence narrowing steadily as more respondents are observed, while remaining valid to inspect at any point along the way – there is no multiple-looks penalty for having watched, say, testimony consistency firm up over the first thousand or so respondents before checking in on country of origin or religion.

4.2 e-values over time

Figure 2 shows the corresponding e-value trajectories. Every one of the nine curated e-values exceeds the $1/\alpha$ rejection threshold by the end of the sample. The number of respondents needed to first cross that threshold varies enormously across attributes: from 405 for *Reason for migrating: Ethnic persecution* up to 5,805 for *Country of origin: Syria* – a much wider range than the camera case study's 25-110, reflecting both weaker effects and a far larger $g^\star$ here, which shrinks early evidence more aggressively. As in the camera study, crossing need not be permanent: 2 of the nine curated e-values (*Gender: Male* and *Country of origin: Syria*) dip back below the threshold at least once after their first crossing. This is expected of e-processes and is harmless under the relevant decision rule – stop and reject the first time the threshold is crossed, rather than requiring it to stay crossed forever.

Not shown in Figure 2, but visible in the full comparison below: the curated nine are the *strongest* AMCE per attribute by construction, and not every one of the 27 AMCEs in this study reaches the rejection threshold, even with the full sample.

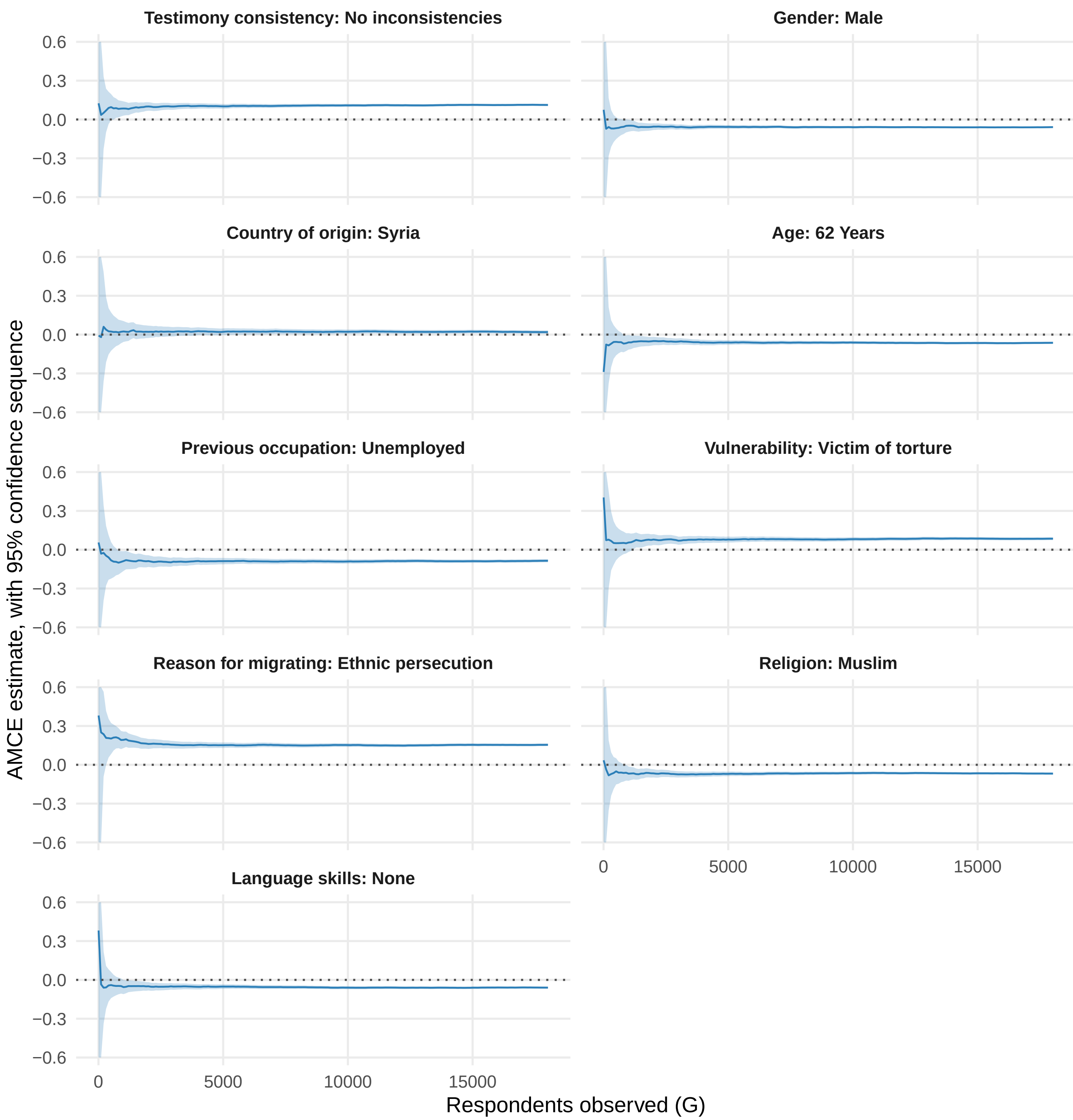

Figure 1: Anytime-valid 95% confidence sequence for the nine curated AMCEs (one per attribute; see Method) as respondents accumulate. Shaded regions are clipped at ± 0.6, which is wider than every finite bound in the data (the widest is 0.565 in absolute value) – so the clip affects only genuinely-infinite early bounds, never a legitimate finite one. Below $G = 205$, the confidence-sequence radius is infinite for *every* one of the 27 coefficients in this dataset, curated or not, since the radius depends only on G, $g^\star$, and α – not on the coefficient's own estimate – and is expected: a confidence sequence must sometimes be uninformative to guarantee valid coverage at every sample size, not just a single pre-planned n.

Figure 2: Anytime-valid e-value for the same nine curated AMCEs as respondents accumulate (log scale). The dashed red line is $1/\alpha = 20$, the threshold an e-value must cross to reject the null of no effect at level α; the dotted grey line is $e = 1$ (neutral evidence). e-values are capped at 10^{300} for display, which affects only Reason for migrating: Ethnic persecution: from $G = 14,805$ its true e-value exceeds 10^{300}, and from $G = 15,205$ onward it exceeds the range representable in double-precision floating point altogether (i.e., is exactly infinite); both are shown at the display cap. The underlying CSV retains the true, uncapped (∞) values throughout.

4.3 Fixed-n interval vs. confidence sequence, at the final sample size

Figure 3 makes explicit the price of anytime validity at a single, fixed sample size: the two methods share the same point estimate (they are computed from the same model fit), but the confidence sequence is visibly wider than the conventional interval for every AMCE. Concretely, both intervals exclude zero for 20 of the 27 AMCEs, and for none does the anytime-valid interval exclude zero while the conventional interval does not – consistent with the anytime-valid interval never being narrower here. But 4 AMCEs (*Country of origin: Kosovo; Country of origin: Ukraine; Age: 38 Years; Vulnerability: No surviving family*) are flagged as distinguishable from zero by the conventional fixed-n interval while the anytime-valid confidence sequence still contains zero, and a further 3 (*Country of origin: Eritrea; Country of origin: Iraq; Country of origin: Pakistan*) are indistinguishable from zero under either method. That extra width purchased by the anytime-valid interval is exactly what buys the ability to monitor the study continuously, as in Figures 1 and 2, and still report a valid interval or rejection decision at whatever point the analyst chooses to stop – something the conventional interval does not license.

5 Discussion

On this large, real conjoint dataset, the anytime-valid method continues to behave exactly as the parent paper's theory predicts: confidence sequences that are valid to inspect continuously, at the cost of extra width relative to a conventional interval computed once at a fixed sample size. Two features of this particular dataset are worth drawing out because they did not appear in the smaller camera case study.

First, not every AMCE is distinguishable from zero, even with 18,021 respondents. Of the 4 AMCEs that the conventional interval alone flags – Country of origin: Kosovo; Country of origin: Ukraine; Age: 38 Years; Vulnerability: No surviving family – 2 (namely *Country of origin: Kosovo* and *Country of origin: Ukraine*) did cross the anytime-valid rejection threshold at some earlier point in sequential monitoring before falling back below it by the end, illustrating concretely why a rule that only checks significance once, at a pre-committed n, is not the same rule as one that must remain valid at every n an analyst might stop at: an analyst free to stop the moment either of these crossed would have "found" a significant effect under the conventional logic, but that finding would not have survived to the final sample, and the anytime-valid procedure never licenses treating it as a discovery in the first place. 2 more (*Age: 38 Years* and *Vulnerability: No surviving family*) never cross the anytime-valid threshold at all in this dataset, despite the conventional interval excluding zero throughout. A further 3 AMCEs (*Country of origin: Eritrea; Country of origin: Iraq; Country of origin: Pakistan*) are not distinguishable from zero under either method. All of this is a substantive as well as a statistical observation: several country-of-origin effects (relative to the Afghanistan baseline) and one age contrast are genuinely small and hard to pin down even at this scale, in contrast to the uniformly large, easily detected effects of testimony consistency, reason for migration, and vulnerability that dominate Figure 3, echoing the original Bansak et al. (2016) finding that humanitarian concerns dominate European asylum preferences.

Second, this dataset shows what the anytime-valid math looks like at the edge of floating-point range. With 18,021 clusters and some very large effects, several e-values grow large enough to overflow to ∞ in double precision well before the study ends (see Figure 2's caption); the underlying data preserve the true (infinite) values, and only the figure's display is capped. This did not arise in the smaller camera case study and is a reminder that "e-value $= \infty$" is not a bug but the expected behavior of an e-process once evidence for a real, large, and precisely-estimated effect becomes

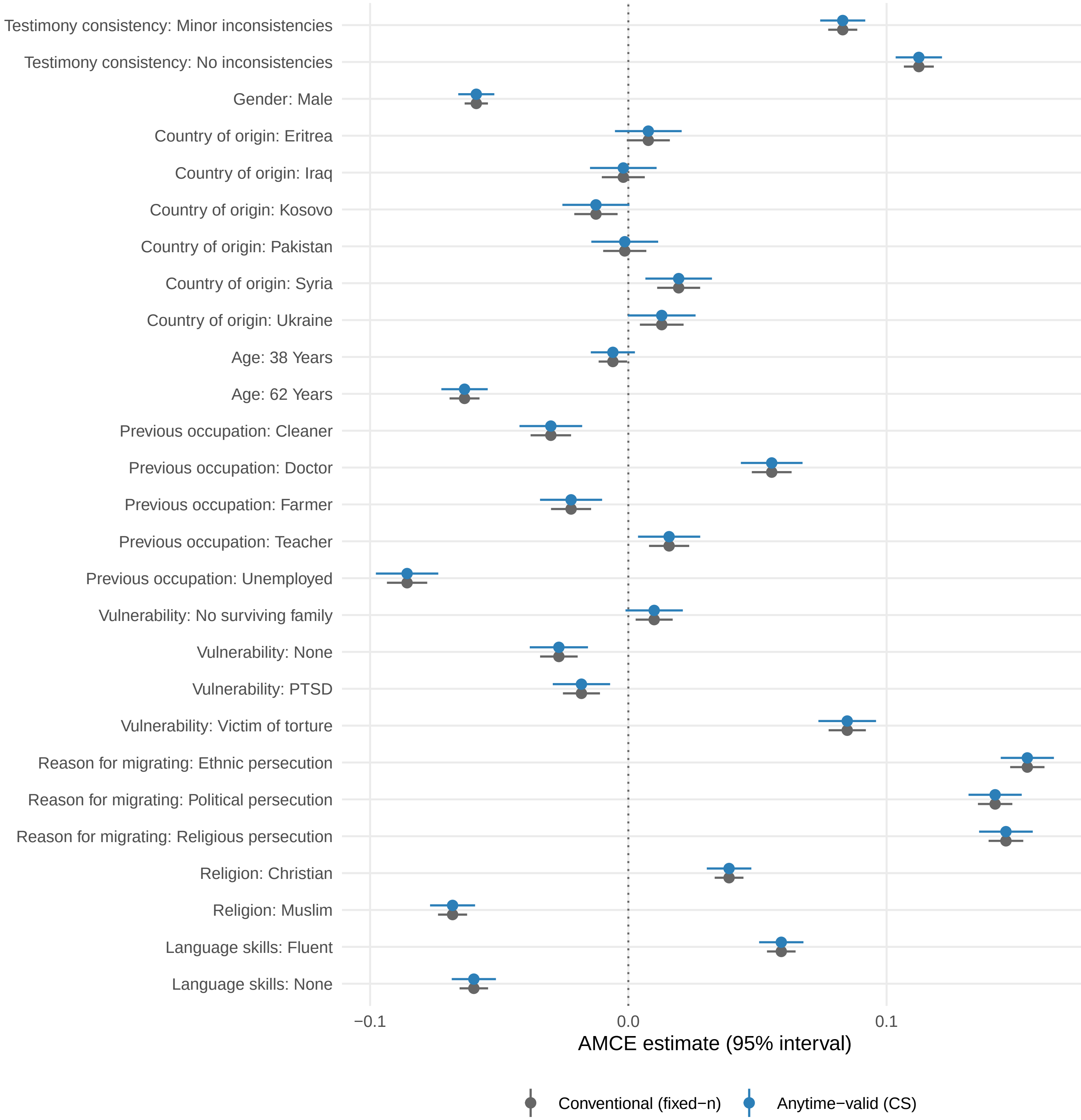

Figure 3: Final AMCE point estimate with a conventional fixed-n 95% confidence interval versus the anytime-valid 95% confidence sequence, for all 27 AMCEs, both computed from the same final model fit ($G = 18,021$).

overwhelming.

6 References

- Bansak, K., Hainmueller, J., and Hangartner, D. (2016). How economic, humanitarian, and religious concerns shape European attitudes toward asylum seekers. *Science*, 354(6309), 217-222.
- Bansak, K., Hainmueller, J., and Hangartner, D. (2023). Europeans' support for refugees of varying background is stable over time. *Nature*, 620(7975), 849-854. Replication data: Harvard Dataverse, doi:10.7910/DVN/FTL1MM.
- Lindon, M., Ham, D., Tingley, M., and Bojinov, I. (2026). Anytime-valid linear models and regression adjusted causal inference in randomized experiments. *Journal of the American Statistical Association.* arXiv:2210.08589.
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